SliderMC Command Cheat Sheet

Firmware V1.0
DIN A4 · ASCII protocol

S — Set (session)

Short	Long	Description
SS	`SetSpeed`	Cruise speed mm/s (≤ max_speed); bare reloads init_speed; live on next fill (incl. MJ). Dual MT: axis1=session, axis2×ratio.
SA	`SetAccel`	Peak accel mm/s² (≤ max_accel); bare reloads init_accel; live on next fill (incl. MJ). Dual MT: same ratio scaling as SS.
SE	`SetEnable`	Driver enable 0|1; bare toggles; required before motion; off = hard stop.
ST	`SetTerminal`	Terminal Mode 0|1; bare toggles; local echo + UART sniff to USB (expert).
SV	`SetVerbose`	Verbose #… push 0|1; bare toggles; ~3 Hz (rate via verbose_rate_hz).
SD	`SetDebug`	USB-only debug level 0..5; bare restores default; never on UIC UART.
SL	`SetLeft`	Session soft min (working window); bare→slider_min; none clears (→envelope if set); skip _; !E:limit past envelope.
SR	`SetRight`	Session soft max; bare→slider_max; none clears; skip _; !E:limit if left>right.

G — Get (session)

Short	Long	Description
GS	`GetSpeed`	Current session cruise speed.
GA	`GetAccel`	Current session acceleration.
GE	`GetEnable`	Driver enable state.
GT	`GetTerminal`	Terminal Mode state.
GV	`GetVerbose`	Verbose push state.
GD	`GetDebug`	USB debug level.
GL	`GetLeft`	Session soft min; effective (session else envelope); - if both None; dual when axis2 on.
GR	`GetRight`	Session soft max; same effective / - rules as GL.

I — Is / status

Short	Long	Description
IM	`IsMoving`	Moving or settling on any active axis.
IH	`IsHoming`	Homing cycle active.
IL	`IsLimit`	At soft-limit position (axis1).
IE	`IsError`	PIN_DRV_ERROR / EMO latched.
IP	`IsPosition`	Axis-1 position; second field when axis2_use=1.
IA	`IsAxis`	Active axis count (config_axis2_enabled).
IT	`IsTarget`	Axis-1 seek target, or - if none / soft-stop.
IR	`IsReady`	1 only if idle, not homing, enabled, and not waiting.
IW	`IsWaiting`	1 if any W / WM / WH wait is active.
ID	`IsDiag`	Motion diag counters (FIFO underrun, peak STEP Hz, …).
IZ	`IsReset`	Last chip reset: power|wdt|run|soft|debug|brownout|…
IX	`Pinout`	ASCII GP / name / desc (≤80 cols). Axis-2 rows only if axis2 on.

M — Movement

Short	Long	Description
MT	`MoveTo`	Absolute user units; optional 2nd axis; skip _; needs SE; live-retarget. Dual: time-sync ratio.
M	`Move`	Relative move (alias MoveBy); same dual/skip rules as MT.
ML	`MoveLeft`	Jog −; mask 0=both, 1=axis1, 2=axis2 when axis2 on; soft-stop MS/!.
MR	`MoveRight`	Jog +; mask same as ML; soft-stop MS/!.
MJ	`MoveJoy`	Joy speed % of SS, signed (− left / + right); 2-axis optional pct2 (omit=0); 0=soft-stop; SS/SA live; clamp max_speed[_2].
MH	`MoveHome`	Homing; axis 1 (default) or 2; no-op if home_mode=0; cancel MS/H.
MS	`MoveStop`	Soft decelerate both axes; keeps enable; ends joy-mode; does not cancel waits. Dual: scaled accel kept.

P — Path

Short	Long	Description
PC	`PathClear`	Clear path buffer (count→0); !E:busy while PG active.
PD	`PathData`	Append signed µm sample(s); optional axis2; skip _ →0; OK while PG (live stream).
PG	`PathGo`	Play buffer from sample 0; needs SE; !E:empty|busy|disabled. MS/H ends path.
PN	`PathNumber`	Samples in buffer; allowed during PG.
PS	`PathSlice`	Slice length µs (≥1000); bare→init_path_slice_us; !E:busy while PG.

X — Extender

Short	Long	Description
X0-3	`Ext0-3`	Ext out n logical 0|1; bare toggles; glued X00≡X0 0; OK during EMO. X4+ rejected.

C — Config

Short	Long	Description
CS	`ConfigSet`	Persist key to mc.ini; silent ok. axis2_use / WDT_use need RB to take HW effect.
CR	`ConfigReset`	Reset all config to compiled defaults and save mc.ini.
CG	`ConfigGet`	One key, or bare dumps all keys (multi-line).
RB	`Reboot`	Soft MCU reset (no power cycle); EN off first. After CS axis2_use.

W — Wait

Short	Long	Description
W	`Wait`	Delay then continue ; chain; bare→1 s; never !E:timeout.
WM	`WaitMoving`	Pause chain until move ends; optional timeout → !E:timeout, cancel rest of chain.
WH	`WaitHoming`	Pause until homing ends; timeout same as WM.

V — Version

Short	Long	Description
VA	`VersionAbout`	About string (name, version, author).
VF	`VersionFW`	Firmware version.
VP	`VersionProtocol`	Protocol version.

Special

Short	Long	Description
H/HT	`Halt`	Immediate STEP abort; enable off; cancel waits and remaining ; chain.
P	`Pins`	Machine-readable pin map (alias VersionGPIO). Axis-2 pins if axis2 on.
$/HL	`Help`	ASCII table of all commands (≤80 columns).

Wire: one command per line (`\n`); chain with `;`; `#` comment to EOL (comment-only lines ignored); bare bool setters toggle; motion/settings silent on success; errors `!E:code message`.
Realtime (no newline): `?` status · `!` soft stop · `Ctrl-X` (0x18) soft reset.
Status (`#X …`): `I` idle · `A` accel · `M` cruise · `B` decel · `H` homing · `P` path · `L` hard-limit · `D` disabled · `E` error. Moving: `#M/#A/#B pos speed accel [target]`.
Halt vs Stop: `MS`/`!` soft decel (enable kept, ends joy-mode); `H`/`HT` immediate abort, enable off, cancel waits. `MJ`: skip if value unchanged.